Beyond Pass@1 RL

Lab Meeting

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Motivation

- For a policy π that samples trajectories τ with reward $r(\tau)$, define

$$\text{best@}k(\pi) = \mathop{\mathbb{E}}_{\tau_1,\ldots,\tau_k\sim\pi}[\max_i r(\tau_i)]$$

- $\text{pass@}k(\pi)$ is $\text{best@}k$ restricted to binary rewards $r \in \{0, 1\}$. We'll use the two interchangeably.
- We measure $\text{pass@}k$ over a distribution of problems, not a single problem, so in general $\text{pass@}k \neq 1 - (1 - \text{pass@}1)^k$. The two are equal for a single problem.

Motivation

- Naive RL with verifiable rewards (RLVR) improves $\mathrm{pass@1}$ and $\mathrm{pass@}k$ for small k, but at large k the RL-trained model is worse than the base model (Y. Yue et al. 2025).
- For a fixed prompt, $\mathrm{pass@}k$ is monotone in $\mathrm{pass@1}$, so RLVR is actually worse than the base model on harder problems.
- So we want RL that raises $\mathrm{pass@}k$, not just $\mathrm{pass@1}$: solving harder problems and exploring more. But what should that mean as a training objective?

What Does Raising Pass@k Mean?

1. **Harder problems.** Put more weight on problems the model rarely solves: $\mathrm{pass@}k$ objectives, MaxRL, curricula. With binary rewards, these all just reweight prompts.
2. **Diversity within a problem.** With one fixed reward, the best $\mathrm{pass@}k$ policy is deterministic. Reward uncertainty and set-level objectives credit each sample for what it adds to a set.

Optimizing $\mathrm{pass@}k$ Directly

One answer: optimize $\mathrm{pass@}k$ directly (Walder and Karkhanis 2025).

Setup. For a fixed prompt, sample $n \geq k$ i.i.d. trajectories $\tau_1, \ldots, \tau_n \sim \pi_\theta$.

- $r_i = r(\tau_i) \in \{0, 1\}$ indicates whether τ_i is correct.
- $c = \sum_i r_i$ is the number of correct samples.

We write $\widehat{\nabla} X$ for an unbiased estimator of $\nabla_\theta X$. The standard policy-gradient (REINFORCE) estimator is

$$\widehat{\nabla}\mathrm{pass@1} = \frac{1}{n}\sum_{i=1}^{n} r_i \nabla_\theta \log \pi_\theta(\tau_i), \qquad \mathbb{E}[\widehat{\nabla}\mathrm{pass@1}] = \nabla_\theta \mathrm{pass@1}.$$

It only optimizes $\mathrm{pass@1}$.

Pass@k Policy Optimization (PKPO)

With τ_i, r_i, c as before, PKPO uses the unbiased $\mathrm{pass@}k$ estimator

$$\rho(n, c, k) = 1 - \frac{\binom{n-c}{k}}{\binom{n}{k}}.$$

Reweighting each sample by w_i gives an unbiased estimator of $\nabla_\theta \mathrm{pass@}k$:

$$\widehat{\nabla}\mathrm{pass@}k = \sum_{i=1}^{n} w_i \nabla_\theta \log \pi_\theta(\tau_i), \qquad w_i = \begin{cases} \frac{k}{n} & \text{if } r_i = 1 \\ \frac{k}{n}\,\rho(n-1, c, k-1) & \text{if } r_i = 0 \end{cases}$$

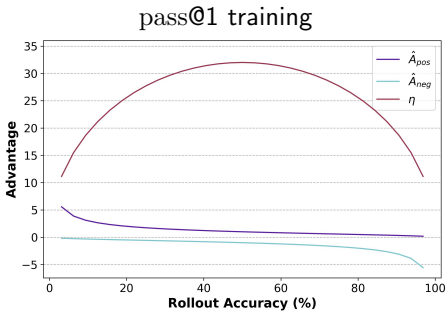
- The advantage of successes on rarely solved problems is much higher.
- At $k = 1$, PKPO reduces to REINFORCE.

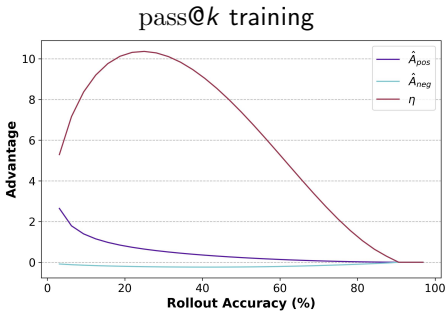
Pass@k Training

Chen et al. (2025) independently derive the same per-sample weights as PKPO (up to a baseline and GRPO-style standardization), and study what $\mathrm{pass@}k$ training does in practice.

- **Shifts focus to harder prompts.** The total weight per prompt peaks at about 25% accuracy in their setting, vs. 50% for GRPO, so training concentrates nearer the edge of competence.
- **Keeps exploring.** Policy entropy stays high under $\mathrm{pass@}k$ training but collapses under $\mathrm{pass@1}$ training.
- **Transfers to $\mathrm{pass@1}$.** Training on $\mathrm{pass@}k$ first, then switching to $\mathrm{pass@1}$, beats $\mathrm{pass@1}$ training alone (Qwen2.5-7B on the Enigmata logic puzzles: $\mathrm{pass@1}$ of 30.8 vs. 12.9, from 4.7 for the base model).

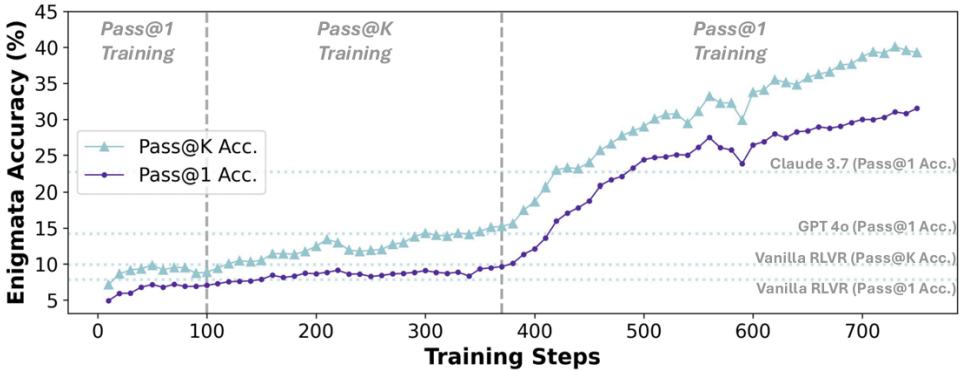
Pass@k Training: Where the Gradient Goes





$\eta = \sum_{i=1}^{n} |w_i|$ is the total weight a prompt receives, plotted against the prompt's pass@1 (x-axis).

Pass@k Training: Results



Accuracy on Enigmata over training. Vanilla RLVR uses pass@1 reward throughout.

Maximum Likelihood RL (MaxRL)

Tajwar et al. (2026): let $p = \mathrm{pass@1}$ on a given prompt. Over the distribution of prompts, RL maximizes $\mathbb{E}[p]$, while maximum likelihood (ML) maximizes $\mathbb{E}[\log p]$:

$$\nabla_\theta \, \mathbb{E}[\log p] = \mathbb{E}\left[\frac{1}{p} \nabla_\theta p\right],$$

so ML puts far more weight on prompts the model rarely solves. Expanding $\log p$ as a series in $1 - p$:

$$\nabla_\theta \log p = \sum_{k=1}^{\infty} \frac{1}{k} \nabla_\theta \mathrm{pass@}k.$$

MaxRL: Gradient Estimator

MaxRL truncates this series at $k = n$, the number of samples per prompt, giving an objective $J^{(n)}$ with $\nabla_\theta J^{(n)} = \sum_{k=1}^{n} \frac{1}{k} \nabla_\theta \mathrm{pass@}k$.

With τ_i, r_i, c as before, its estimator averages $\nabla_\theta \log \pi_\theta$ over *correct samples only*:

$$\widehat{\nabla} J^{(n)} = \begin{cases} \frac{1}{c} \sum_{i=1}^{n} r_i \nabla_\theta \log \pi_\theta(\tau_i) & \text{if } c \geq 1 \\ 0 & \text{if } c = 0 \end{cases} \qquad \mathbb{E}[\widehat{\nabla} J^{(n)}] = \nabla_\theta J^{(n)}.$$

- MaxRL differs from REINFORCE only in dividing by c instead of n.
- More samples raise the order of the objective, rather than only reducing variance.

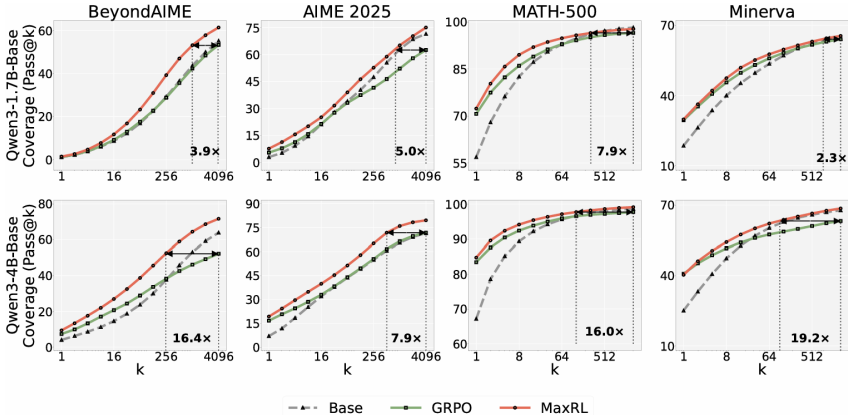
Comparing Gradient Estimators

With τ_i, r_i, c as before, all three estimators have the form $\sum_{i=1}^{n} w_i \nabla_\theta \log \pi_\theta(\tau_i)$ and differ only in the per-sample weight w_i. Highlighted: changes from REINFORCE.

Method	Weight w_i	Unbiased for
REINFORCE / GRPO	r_i/n	$\nabla_\theta \mathrm{pass@1}$
PKPO	$\begin{cases} k/n & \text{if } r_i = 1 \\ k\rho(n-1, c, k-1)/n & \text{if } r_i = 0 \end{cases}$	$\nabla_\theta \mathrm{pass@}k$
MaxRL	$r_i/c \quad (0 \text{ if } c = 0)$	$\nabla_\theta J^{(n)}$

Baselines (e.g. leave-one-out), standardization, and GRPO's clipping for off-policy updates are omitted from all rows; with these omitted, GRPO (Shao et al. 2024) is REINFORCE.

MaxRL: Results



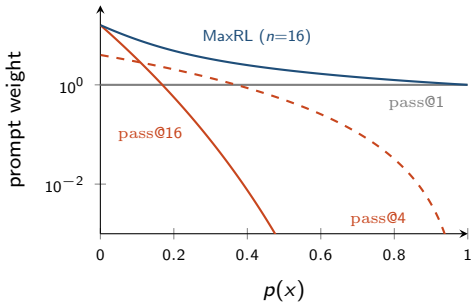
MaxRL beats GRPO across the full $\mathrm{pass@}k$ spectrum, and matches or exceeds the base model at ultra-high k.

Critique: Pass@k Only Reweights Prompts

Yu (2025) and Barakat et al. (2026): with binary rewards,

$$\nabla_\theta \mathrm{pass@}k(x) = k\,(1-p(x))^{k-1}\,\nabla_\theta p(x).$$

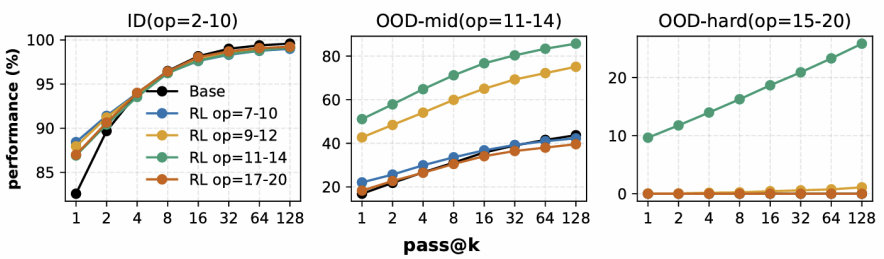
- With a baseline, every estimator in the comparison table updates a prompt along $\sum_i (r_i - c/n)\nabla_\theta \log \pi_\theta(\tau_i)$. They differ only in prompt weight.
- So $\mathrm{pass@}k$ training is an implicit curriculum, and it gives no signal when $c = 0$.



RL at the Edge of Competence

Zhang, Neubig, and X. Yue (2026) suggest that the best problems to train on have low $\mathrm{pass@1}$ (under 20%), well below the 50% that GRPO focuses on.

Controlled setting: synthetic arithmetic whose difficulty is the number of operations (op), and a 100M-parameter model pre-trained only on op ≤ 10. RL is run on one difficulty range (colors), and $\mathrm{pass@}k$ is evaluated on three test ranges (panels).



Liu et al. (2025) see the same pattern at scale (1.5B model, 136K problems): RL gains are largest on tasks where the base model is weakest.

Reweighting Is Not Diversity

- So far, raising $\mathrm{pass@}k$ has meant putting more weight on hard problems, through the objective or the data.
- But with a single fixed reward, $\mathrm{pass@}k$ and $\mathrm{best@}k$ always have a *deterministic* optimal policy: put all mass on one best answer (GX-Chen et al. 2026).
- To reward diversity, we need set-level rewards that make the optimal policy stochastic.

Randomized Objectives, Set Actions (ROSA)

GX-Chen et al. (2026): treat the reward as uncertain. Let $\mathcal{R}$ be a distribution over reward functions $\tilde{r}$ (e.g. different user preferences, or an ensemble of noisy judges), and maximize

$$J_{\mathrm{ROSA}} = \mathop{\mathbb{E}}_{\tau_1,\ldots,\tau_n \sim \pi_\theta} \mathop{\mathbb{E}}_{\tilde{r}\sim\mathcal{R}} \left[\max_i \tilde{r}(\tau_i) \right],$$

i.e. $\mathrm{best@}n$ averaged over reward functions. The optimal policy is now stochastic: it covers the answers that are best under each $\tilde{r}$, in proportion to that $\tilde{r}$'s probability.

ROSA: Gradient Estimator

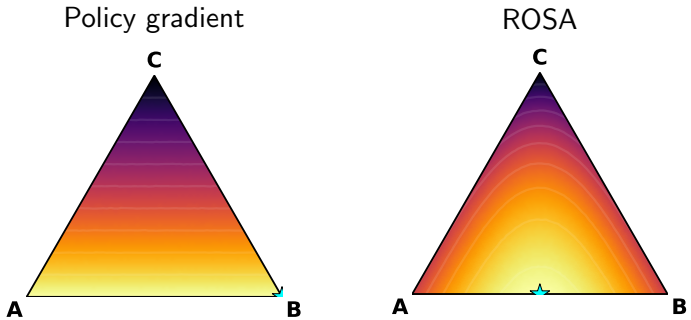
Like the estimators on the comparison slide, ROSA's gradient estimator has the form $\sum_{i=1}^{n} w_i \nabla_\theta \log \pi_\theta(\tau_i)$. The weights are

$$w_i = \mathop{\mathbb{E}}_{\tilde{r}\sim\mathcal{R}} \left[\max_j \tilde{r}(\tau_j) - \max_{j\neq i} \tilde{r}(\tau_j) \right],$$

estimated in practice by averaging over a few sampled $\tilde{r}$.

- $w_i > 0$ only if τ_i is the best sample under some $\tilde{r}$; every other sample gets zero weight.
- With a single fixed reward ($\tilde{r} = r$ always), this is an unbiased estimator of $\nabla_\theta \mathrm{best@}n$.

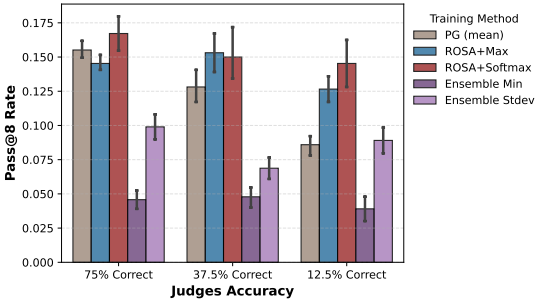
ROSA: Toy Example

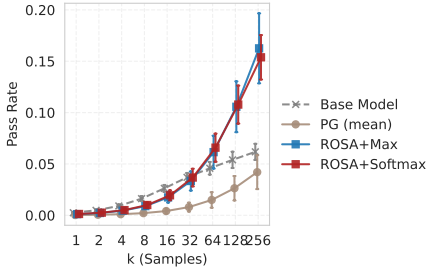


Each point is a policy over three actions; brighter means a higher expected objective ($n = 4$ samples). Two equally likely reward functions: $\tilde{r}_A$ rewards only A, $\tilde{r}_B$ only B.

- PG is maximized anywhere on the A–B edge: nothing prefers mixing A and B.
- ROSA has a unique maximum at 50/50 (star).

ROSA: Noisy Judges





Gemma 2 2B on MATH. Each problem has 8 judges, of which only some (6, 3, or 1) reward the correct answer; the learner doesn't know which. ROSA+Softmax is a smoothed variant of ROSA.

- **Left** (in distribution, $\mathrm{pass@8}$): ROSA degrades much less than PG as judges get worse. (Ensemble Min/Stdev: pessimistic aggregates of the judges' rewards.)
- **Right** (AIME 2025, out of distribution): PG falls *below the base model* at large k; ROSA rises above it.

Vector Policy Optimization (VPO)

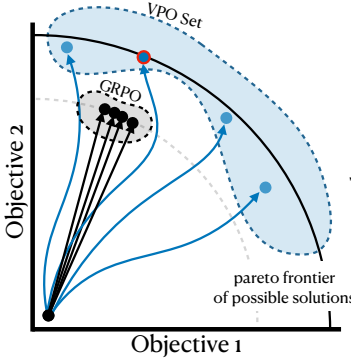
Bahlous-Boldi et al. (2026): for models used inside test-time search, train for a *diverse set* of good solutions.

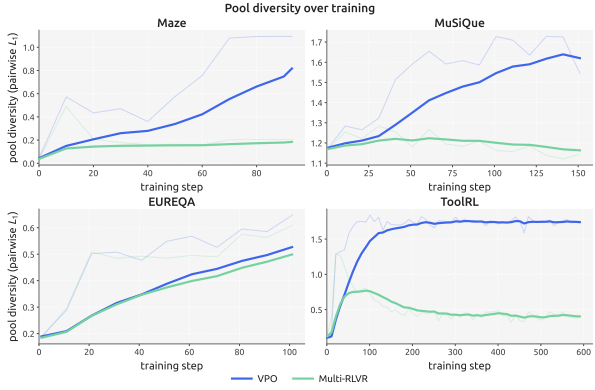
- Vector reward $\mathbf{r}(\tau) \in \mathbb{R}^D$, e.g. pass/fail on each of D test cases.
- One rollout writes a set of m answers $S = \{\tau_1, \ldots, \tau_m\}$, scored by

$$R(S) = \mathop{\mathbb{E}}_{w \sim \mathrm{Dir}(\mathbf{1})} \left[\max_{\tau \in S} w^\top \mathbf{r}(\tau) \right],$$

  with w a uniformly random weighting ($w \geq 0$, $\sum_j w_j = 1$).
- Looks like ROSA with $\tilde{r} = w^\top \mathbf{r}$, but it's just GRPO with reward $R(S)$: no per-answer credit.

VPO: Results





- **Pool diversity:** mean L_1 distance $\|\mathbf{r}(\tau_i) - \mathbf{r}(\tau_j)\|_1$ between reward vectors of answers in a set.
- **Multi-RLVR** (Puri et al. 2026): m-answer (non-i.i.d.) rollouts with a fixed weighting w^* and a sum over answers: $\sum_{\tau \in S} {w^*}^\top \mathbf{r}(\tau)$, vs. VPO's $\mathbb{E}_w[\max_{\tau \in S} w^\top \mathbf{r}(\tau)]$.

Polychromic Exploratory Policy Optimization (Poly-EPO)

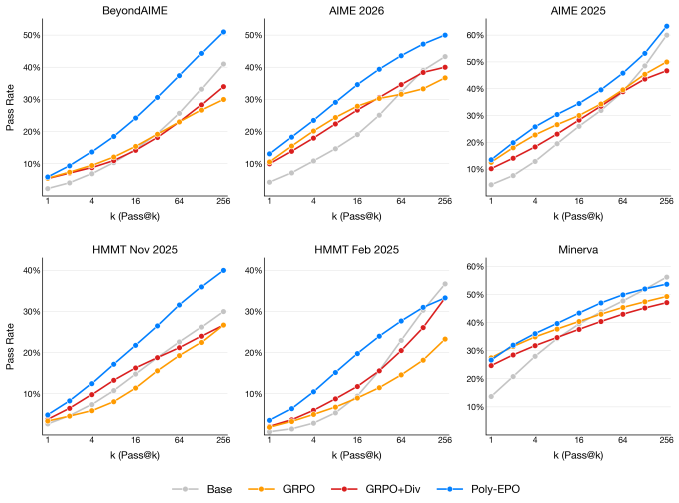
Orney et al. (2026): also scores sets of answers, but from i.i.d. samples and with per-answer credit.

- Score a set S of m answers by mean reward times strategy diversity:

$$f(S) = \left(\frac{1}{m}\sum_{\tau \in S} r(\tau)\right) \cdot \frac{|\{C(\tau) : \tau \in S\}|}{m},$$

  where an LM judge clusters answers by reasoning strategy $C(\tau)$.
- Sample n i.i.d. answers. Answer τ_i's advantage is $\mathbb{E}_{S \ni \tau_i}[f(S)] - \mathbb{E}_S[f(S)]$, over size-$m$ subsets S of the samples.
- An incorrect answer can get positive advantage if it adds a new strategy to sets that also contain correct answers.

Poly-EPO: Results



- Qwen3-4B-Base on POLARIS-53k math; $n = 8$ samples, sets of $m = 4$.
- GRPO+Div: GRPO with a per-answer diversity bonus from the same clustering.
- The base model overtakes GRPO and GRPO+Div as early as $k = 32$. Poly-EPO stays above the base model at $k = 256$ on 4 of 6 benchmarks.

Marginal Set-Level Contributions

Different ways to credit individual answers for their contribution to a set-level objective $f_{\tilde{r}}(S)$, averaged over reward functions $\tilde{r} \sim \mathcal{R}$.

Leave one out:

$$w_i = \mathop{\mathbb{E}}_{\tilde{r}} \left[f_{\tilde{r}}(\tau_{1:n}) - f_{\tilde{r}}(\tau_{1:n} \setminus \tau_i) \right]$$

Set size is tied to the group size n.

Enumerate or sample subsets:

$$w_i \propto \mathop{\mathbb{E}}_{\tilde{r}} \left[\mathop{\mathbb{E}}_{S \ni \tau_i} \left[f_{\tilde{r}}(S) \right] - \mathop{\mathbb{E}}_{S' \not\ni \tau_i} \left[f_{\tilde{r}}(S') \right] \right], \quad |S| = |S'| = m \leq n$$

Any set size, and averaging over many sets lowers variance.

Big Ideas

- Measuring the full $\mathrm{pass@}k$ spectrum allows us to see how RL affects problems of varying difficulties.
- "Raising $\mathrm{pass@}k$" can mean two things: more weight on hard problems, or more diversity within a problem.
- Naive REINFORCE/GRPO training gives little gradient to hard problems. Changing the objective or the data mixture fixes this, and with binary rewards these are the same knob: reweighting prompts.
- Objectives should be designed at the set level rather than the answer level to produce stochastic optimal policies.

Personal Thoughts

- With binary rewards, $\mathrm{pass@}k$ training is a curriculum in disguise. As far as I can tell, nobody has explicitly compared $\mathrm{pass@}k$ methods to curriculum-based baselines.
- Multi-reward benchmarks and tasks are still quite poor, and there is a lot of room for work on set-level objectives.

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